

Problem statement 4:

You are required to create a workflow configuration file (.github/workflow/ci.yml) that perform the following tasks:

1. Checks out the sources code from the repository
2. Sets up the appropriate java Development kit version required for the project
3. Install all necessary project dependencies using Maven.
4. Compiles the application and executes unit tests using appropriate maven commands.
5. Archives the generated build artifacts(such as JAR or WAR files) for future use.

name: Java CI with Maven

on:

push:

branches: ["main"] # Run workflow when code is pushed to main branch

pull_request:

branches: ["main"] # Run workflow when a PR is created to main

jobs:

build:

runs-on: ubuntu-latest # Use a Linux virtual machine to run the pipeline

steps:

Step 1: Get the code from your GitHub repository

- name: Checkout repository

uses: actions/checkout@v4 # Needed so workflow can access your project files

Step 2: Install Java (JDK)

- name: Set up JDK

uses: actions/setup-java@v4

with:

java-version: '21' # Specifies Java version required for project

distribution: 'temurin' # Stable OpenJDK distribution

cache: maven # Caches dependencies to make builds faster

- name: Cache Maven packages

Uses: actions/cache@v4

With:

Path: ~/.m2

Key: \${{runner.os}}-m2-\${{hashFiles('**/pom.xml')}}

Restore-key: |

\${{runner.os}}-m2

Step 3: Build the project

- Name: Build with Maven

run: mvn -B clean install

mvn → runs Maven build tool

-B → batch mode (no interactive input, required for CI)

clean → deletes old compiled files to avoid conflicts
install → compiles code, runs tests, and stores build in local repo

Step 4: Run tests separately

- name: Run Tests

run: mvn test

Ensures all unit tests pass

Helps catch bugs before deployment

-name: Upload Artifact

Uses: actions/upload-artifact@v4

With:

Name: java-build-artifact

Path: target/*.jar